

EE 527P: Practicum on Analysis and Design of Power Electronic Converters

Course Code	EE 527P
Course Name	Practicum on Analysis and Design of Power Electronic Converters
L-T-P-C	0-0-3-2
Prerequisites	EE309 and EE309P- Power Electronics or Equivalent
Intended for	BTech final year and M.Tech in Power Electronics and Drives (PED)
Distribution	Core for M.Tech. in PED, Elective for final year Electrical Engineering and other PG
Approval	17th BoA (May 2017); 13th Senate (Jun 2017)

Course Contents

This is a laboratory course with 3-hour sessions per week. Following is the list of course modules and experiments.

- **Part I** Part-1 will involve experiments related to analysis, design, control and applications' of power electronic converters
 - Basic and few advanced AC-DC, DC-AC, DC-DC and AC-AC converters: Analysis and design (11 hours)
 - Closed loop control of power electronics converter (10 hours)
- Demonstration of few power electronics converter applications such as renewable grid interactive converter, power factor correction circuits etc. (6 hours)
- **Part-II** (15 hours) Part-II will involve a design project related to analysis, design and control of power electronic converters

Text Books

1. Lab experimental manuals will be provided for Part-1.

4. References:

1. Mohan N., Undeland T.M. and Robbins W.P., **Power Electronics – Converters, Applications and Design**, 3rd Edition, Wiley India, 2008
2. L. Umammd, **Power Electronics: Essentials & Applications**, Wiley India, 2009.
3. R.W. Erickson, D. Maksimovic, **Fundamentals of Power Electronics**, 2nd Edition, Kluwer Academic Publishers.
4. Bin Wu, **High Power Converters and AC Drives**, IEEE Press, John Wiley & Sons, 2006.
5. Rashid M., **Power Electronics - Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
6. A. T.Pressman, **Switch Mode Power Supply Design**, McGraw-Hill, 1999.
7. R.S. Ramshaw, **Power Electronics Semiconductor Switches**, Chapman & Hall, 1993.
8. D. Grahame Holmes, Thomas A. Lipo, **Pulse width modulation for power converters: principles and practice**, John Wiley & Sons, 2003.